

AIE (UK) Ltd

AIE Power Management Unit (PMU) CAN Interface Control Document (ICD) PMU ↔ Cyber Unit (CYU)

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Table of Contents

Version 1 – Draft	3
1. Sources of Truth	3
2. CYU Liveness and Heartbeat Behavior	3
3. CYU → PMU Messages	4
3.1 CYU_SystemPowerRequirements (0x300)	4
3.2 CYU_TimeSync (0x402)	5
3.3 CYU_TelemetryInputs (0x404)	5
4. PMU → CYU Messages	5
4.1 PMU_DeviceStatus (0x100)	5
4.2 PMU_BatterySummary (0x106)	6
4.3 PMU_PowerOutput (0x200)	6
4.4 PMU_EngineSummary (0x104)	7
4.5 PMU_ThermalSummary (0x400)	7
4.6 PMU_FaultSummary (0x500)	8
5. Fault Code Definitions	9
6. Message Timing and Timeout Expectations	10
7. Normative Language	11
8. PMU State Transition Diagram	11
9. PMU_DeviceStatus – Status Flags and Behaviour	12
10. Status Flag Precedence Rules	12

Version 1

This document defines the CAN interface between the Power Management Unit (PMU) and the Cyber Unit (CYU). It supersedes earlier interface descriptions by separating wire-level definition (DBC) from semantic meaning and behavior.

This ICD, together with PMU_Interface_v1.dbc, provides sufficient detail for third-party flight interface implementation.

1. Sources of Truth

- Battery electrical and thermal data originate from the Battery BMS (e.g. CAN 2.0B or DroneCAN).
- Inverter electrical and thermal data originate from the active inverter (e.g. Sevcon Gen4 or Unitek Bamocar).
- Engine RPM, temperature, and fault state originate from the ECU.
- Fuel level and fuel-low indication originate from CYU or ECU.
- PMU state, inhibits, derating, cooldown sequencing, and fault logic originate from PMU firmware.

2. CYU Liveness and Heartbeat Behavior

The PMU does not use a dedicated heartbeat or “alive” message from the CYU.

Instead:

- The periodic transmission of CYU_SystemPowerRequirements **SHALL be treated as the CYU liveness indicator.**
- Loss of this message **SHALL be interpreted as loss of CYU permission and/or CYU availability.**
- The PMU **SHALL NOT rely on any separate heartbeat message** to determine CYU health.
- The PMU **SHALL** consider CYU_SystemPowerRequirements invalid if not refreshed within the timeout defined in **Section 6**

This design ensures that loss of CYU communication results in automatic withdrawal of power permission.

A separate CYU heartbeat message is intentionally omitted to avoid ambiguity between command validity and liveness.

3. CYU → PMU Messages

3.1 CYU_SystemPowerRequirements (0x300)

Purpose: Express customer intent regarding power usage.

Signal	Values	Meaning	Source of Truth
CYU_Enable	0 / 1	Master permission for PMU operation.	CYU
CYU_ModeRequest	0=IDLE, 1=RUN, 2=CHARGE	Requested operating intent. PMU may delay or refuse.	CYU
CYU_PowerRequest_kW	0...MaxRatedPower_kW	Requested net electrical power. Clamped internally.	CYU
CYU_ResetFaults	0 / 1 (edge)	Request PMU to clear recoverable faults.	CYU
CYU_CmdSeq	0-255 (wrap)	Command sequence counter incremented for each new command transaction	CYU

CYU_ResetFaults SHALL be sampled when a change in CYU_CmdSeq is detected

3.2 CYU_TimeSync (0x402)

Purpose: Provide absolute UTC time for PMU logging.

Signal	Values	Meaning	Source of Truth
CYU_UTC_EpochSeconds	Unix seconds	UTC reference for PMU RTC.	CYU
CYU_UTC_Milliseconds	0–999	Optional sub-second resolution.	CYU

3.3 CYU_TelemetryInputs (0x404)

Purpose: Provide external operational context.

Signal	Values	Meaning	Source of Truth
CYU_FuelLevel_Percent	0–100 %	Estimated remaining fuel.	CYU / ECU
CYU_FuelLow	0 / 1	Fuel low warning. Triggers cooldown sequencing.	CYU / ECU

4. PMU → CYU Messages

4.1 PMU_DeviceStatus (0x100)

Purpose: Report PMU operational state and system viability.

Value	State	Meaning	CYU Expectation
0	OFF	PMU inactive. No power paths closed.	Do not request power
1	PRECHARGE	DC link charging in progress.	Wait
2	READY	PMU ready but idle.	May request power

3	ACTIVE	Power flow active.	Normal operation
4	COOLDOWN	Controlled engine stop.	Do not command stop
5	FAULT	Fault active, power inhibited.	Abort / land

Additional PMU status flags are defined in Section 'PMU_DeviceStatus'

4.2 PMU_BatterySummary (0x106)

Purpose: Abstracted battery health and capability.

Signal	Units	Meaning	Source of Truth
PMU_BatteryVoltage	V	Battery pack voltage.	Battery BMS
PMU_BatteryCurrent	A (signed)	Battery current (+ discharge).	Battery BMS / Inverter
PMU_BatterySOC	%	Estimated state of charge.	Battery BMS
PMU_BatteryFault	0 / 1	Battery fault present.	Battery BMS
PMU_BatteryMaxTemp	°C	Maximum battery temperature.	Battery BMS

4.3 PMU_PowerOutput (0x200)

Purpose: Report actual power flow.

Signal	Units	Meaning	Source of Truth
PMU_ActualPower_kW	kW	Measured net power.	PMU

PMU_PowerDirection	0/1	0 = discharge, 1 = charge.	PMU
PMU_DCLinkVoltage	V	Inverter DC bus voltage.	Inverter

PMU_PowerDirection reflects measured power flow and MAY differ from CYU intent during transients or fault conditions.

4.4 PMU_EngineSummary (0x104)

Purpose: Engine telemetry for flight awareness.

Signal	Units	Meaning	Source of Truth
PMU_EngineRPM	rpm	Engine rotational speed.	ECU
PMU_EngineTemp	°C	Engine temperature.	ECU
PMU_EngineRunning	0 / 1	Engine combustion active.	ECU
PMU_ECUFaultPresent	0 / 1	ECU fault present.	ECU

4.5 PMU_ThermalSummary (0x400)

Purpose: Thermal limits and derating visibility.

Signal	Units	Meaning	Source of Truth
PMU_InverterMaxTemp	°C	Maximum inverter temperature.	Inverter
PMU_ThermalDerateActive	0 / 1	Thermal derating active.	PMU

PMU_ThermalShutdownImminent	0 / 1	Thermal shutdown approaching.	PMU
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4.6 PMU_FaultSummary (0x500)

Purpose: Fault reporting and event correlation.

Signal	Values	Meaning	Source of Truth
PMU_ErrorCounter	0-65535	Increments on each new fault.	PMU
PMU_LastErrorCode	16-bit	Identifier for last fault.	PMU
PMU_LastErrorSource	Enum	PMU / Battery / Inverter / ECU.	PMU
PMU_LastErrorSeverity	Enum	INFO / WARNING / DERATE / FAULT / SHUTDOWN.	PMU

5. Fault Code Definitions

Initial fault code set. This table is extensible and versioned.

Fault Code	Source	Severity	Description	Expected CYU Action
0x0000	NONE	INFO	No fault present	None
0x0101	BATTERY	DERATE	Battery temperature high	Reduce load
0x0102	BATTERY	FAULT	Battery undervoltage	Prepare landing
0x0201	INVERTER	FAULT	Inverter comms missing	Abort mission
0x0202	INVERTER	SHUTDOWN	DC bus overvoltage	Immediate landing
0x0301	ECU	FAULT	Engine overtemperature	Initiate cooldown
0x0302	ECU	SHUTDOWN	Engine overspeed	Immediate shutdown
0x0401	PMU	FAULT	Precharge timeout	Abort start
0x0402	PMU	SHUTDOWN	Internal watchdog reset	Emergency action

6. Message Timing and Timeout Expectations

Recommended update rates and timeout behavior.

Message	Direction	Recommended Rate	Timeout Behaviour
CYU_SystemPowerRequirements	CYU → PMU	≥1 Hz while enabled	PMU inhibits power
CYU_TimeSync	CYU → PMU	1-10 Hz	Logging timestamps stale
CYU_TelemetryInputs	CYU → PMU	≥1 Hz	Fuel treated as unknown
PMU_DeviceStatus	PMU → CYU	≥5 Hz	PMU unavailable
PMU_BatterySummary	PMU → CYU	≥2 Hz	Battery status stale
PMU_PowerOutput	PMU → CYU	≥5 Hz	Power estimate frozen
PMU_EngineSummary	PMU → CYU	≥2 Hz	Engine state stale
PMU_ThermalSummary	PMU → CYU	≥1 Hz	Thermal state stale
PMU_FaultSummary	PMU → CYU	On change + 1 Hz	Fault correlation lost

7. Normative Language

The following terms are used throughout this document in accordance with aerospace interface conventions:

SHALL: Indicates a mandatory requirement.

SHOULD: Indicates a recommended practice.

MAY: Indicates an optional capability or behavior.

8. PMU State Transition Diagram

The following diagram describes the nominal PMU state transitions. Fault conditions may force transitions to FAULT from any state.

OFF

↓ (Power applied)

PRECHARGE

↓ (DC link charged successfully)

READY

↓ (CYU_Enable = 1 and power request present)

ACTIVE

↓ (Fuel low / stop request)

COOLDOWN

↓ (Cooldown complete)

OFF

Any State → FAULT (on unrecoverable fault)

FAULT → OFF (after fault cleared and reset)

9. PMU_DeviceStatus – Status Flags and Behaviour

The following status flags are transmitted as part of the PMU_DeviceStatus message. These flags SHALL be evaluated in conjunction with PMU_State when determining vehicle behaviour and power availability.

Flag	Meaning	CYU Behavioural Expectation
FaultPresent	One or more active faults exist within the PMU, battery, inverter, or ECU.	When asserted, the CYU SHALL assume reduced or unavailable power and SHOULD prepare for termination or landing.
InhibitPresent	PMU is intentionally preventing power delivery despite enable.	When asserted, the CYU SHALL NOT expect power delivery, regardless of PMU_State.
BatteryCommsOK	Battery communications healthy.	If deasserted, the CYU SHOULD assume battery data is stale and operate conservatively.
InverterCommsOK	Inverter communications healthy.	If deasserted, the CYU SHALL treat power delivery as unreliable or unavailable.
ECUCommsOK	ECU communications healthy.	If deasserted, the CYU SHOULD assume engine state is unknown.
CooldownActive	PMU managing controlled engine shutdown sequencing.	When asserted, the CYU SHALL NOT command immediate engine stop unless safety requires it.

10. Status Flag Precedence Rules

1. PMU_State = FAULT SHALL override all other indicators. Power delivery is unavailable.
2. InhibitPresent = 1 SHALL be treated as power unavailable regardless of PMU_State.
3. Loss of inverter communications SHALL be treated as power unavailable.

4. Loss of battery communications **SHOULD** be treated as degraded operation.
5. CooldownActive does not indicate a fault; it represents intentional, managed shutdown behaviour.